PCB DESIGN ASSIGNMENT REPORT

KiCad EDA Schematic & Layout Design Project

Project & Student Information

Student Name:	Rajat Pathak
Branch / Sec:	ECE / C
Assignment Title:	Emergency Panic Alarm System PCB Design (Electronic Doorbell)
Software Used:	KiCad E.D.A. (Version 10.0.7)

Assignment Objective: Design a complete Printed Circuit Board (PCB) for an emergency panic alarm/doorbell circuit in KiCad. The process involves schematic creation, footprint assignment, component layout, track routing, and BOM extraction.

Theory & Working Principle

> **Project Scenario:** A company requires an Emergency Panic Alarm System for office safety. When an employee presses a panic button during an emergency, the system immediately triggers a high-decibel buzzer and a red warning LED to alert nearby personnel.

1. Circuit Overview

The circuit is designed as an automated emergency notification unit operating on a standard 5V DC supply. It combines visual (LED) and acoustic (Buzzer) signaling mechanism controlled via a low-side NPN transistor switch architecture.

2. Key Hardware Components

- **Panic Button (SW1):** Normally-open momentary push button serving as the primary trigger input.
- **Pull-Down Resistor (R2 - 10kΩ):** Holds the transistor base at a stable ground (0V) potential when the switch is unpressed, preventing false triggers.
- **Base Current-Limiting Resistor (R3 - 1kΩ):** Limits current entering the NPN transistor base when SW1 is actuated.
- **NPN Transistor Switch (Q1):** Acts as a solid-state current switch to drive high-current output devices (buzzer and LED) from the input signal.
- **Piezo Buzzer (BZ1):** Provides audio warning indication upon transistor conduction.
- **Red LED (D1) & Current Limiter (R1 - 330Ω):** Provides visual alert indication; R1 prevents excessive forward current through the LED.
- **Smoothing Capacitor (C1 - 10µF):** Filters supply voltage ripple and maintains localized DC stability.

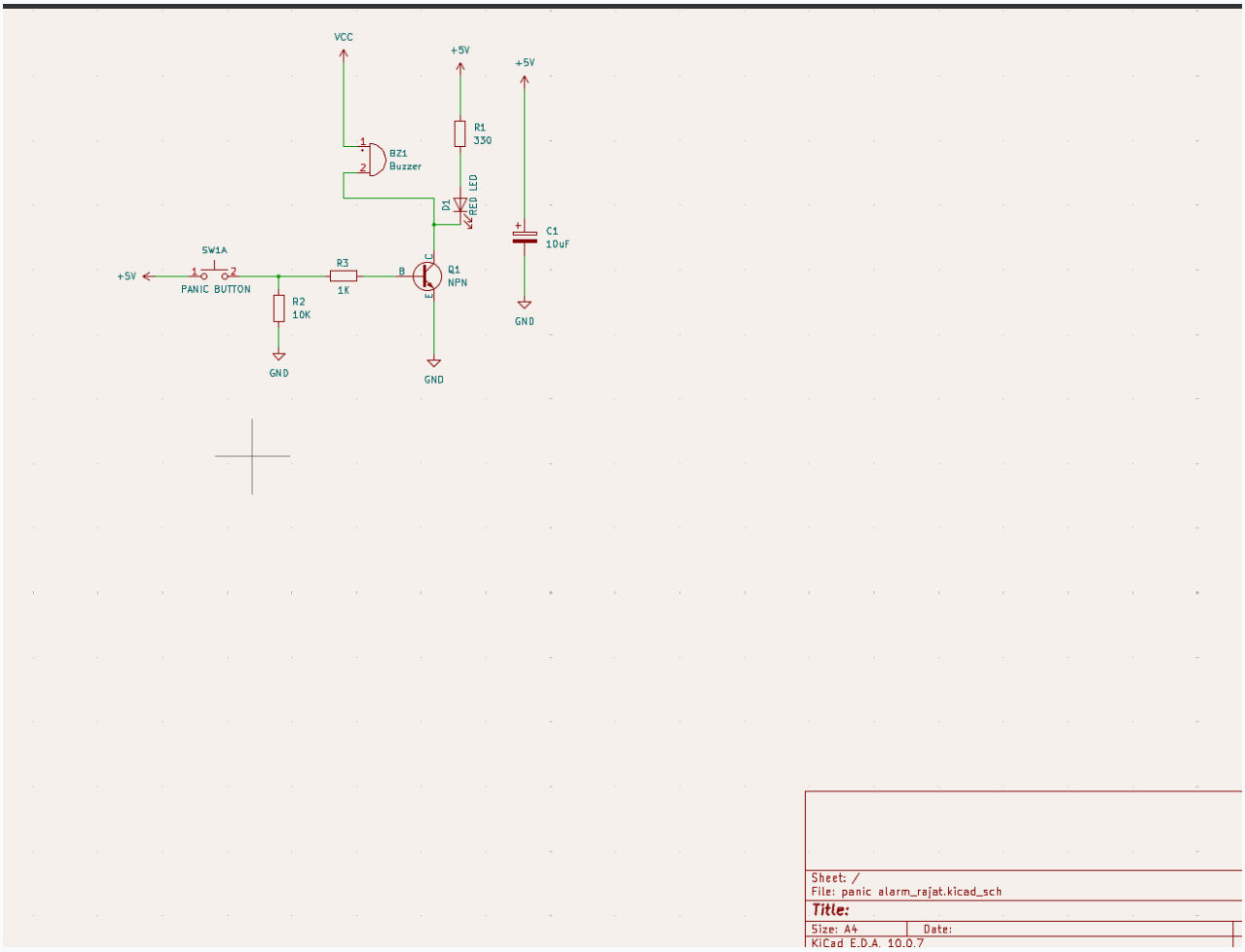
3. Detailed Working Principle

Standby State: When the panic button (SW1) is open, resistor R2 pulls transistor Q1's base to Ground (0V). Transistor Q1 remains in the cutoff state (OFF), drawing zero collector current. As a result, both the Piezo Buzzer (BZ1) and Red LED (D1) remain unpowered and inactive.

Emergency Active State: When an operator depresses SW1, +5V is applied directly across the R3-R2 voltage network, supplying base current to Q1. The transistor rapidly transitions from cutoff into full saturation (ON). This connects the negative terminals of both BZ1 and D1 directly to Ground, allowing current to flow from +5V through the buzzer and LED branch concurrently. The buzzer emits a continuous alarm sound while the Red LED illuminates sharply, signaling an active emergency state until SW1 is released.

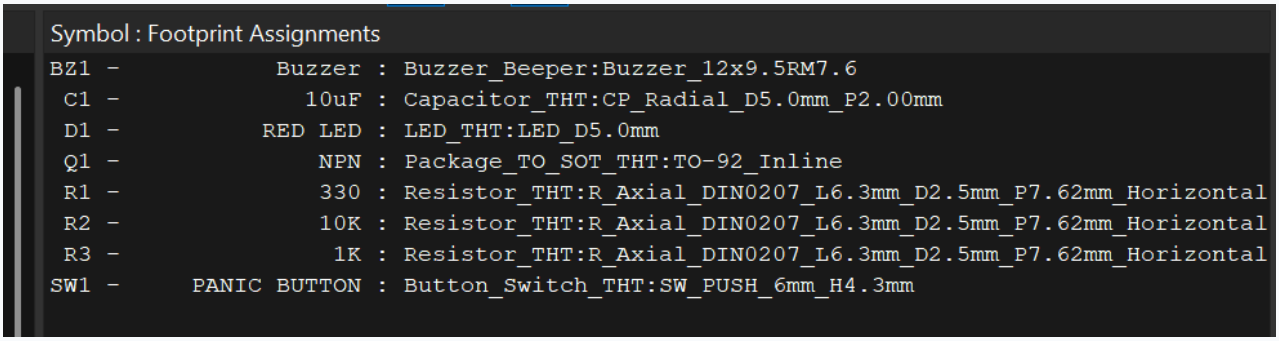
KiCad Schematic Design

The circuit schematic designed in KiCad Schematic Editor showing signal nets, components, and electrical connections:



Footprint Assignment

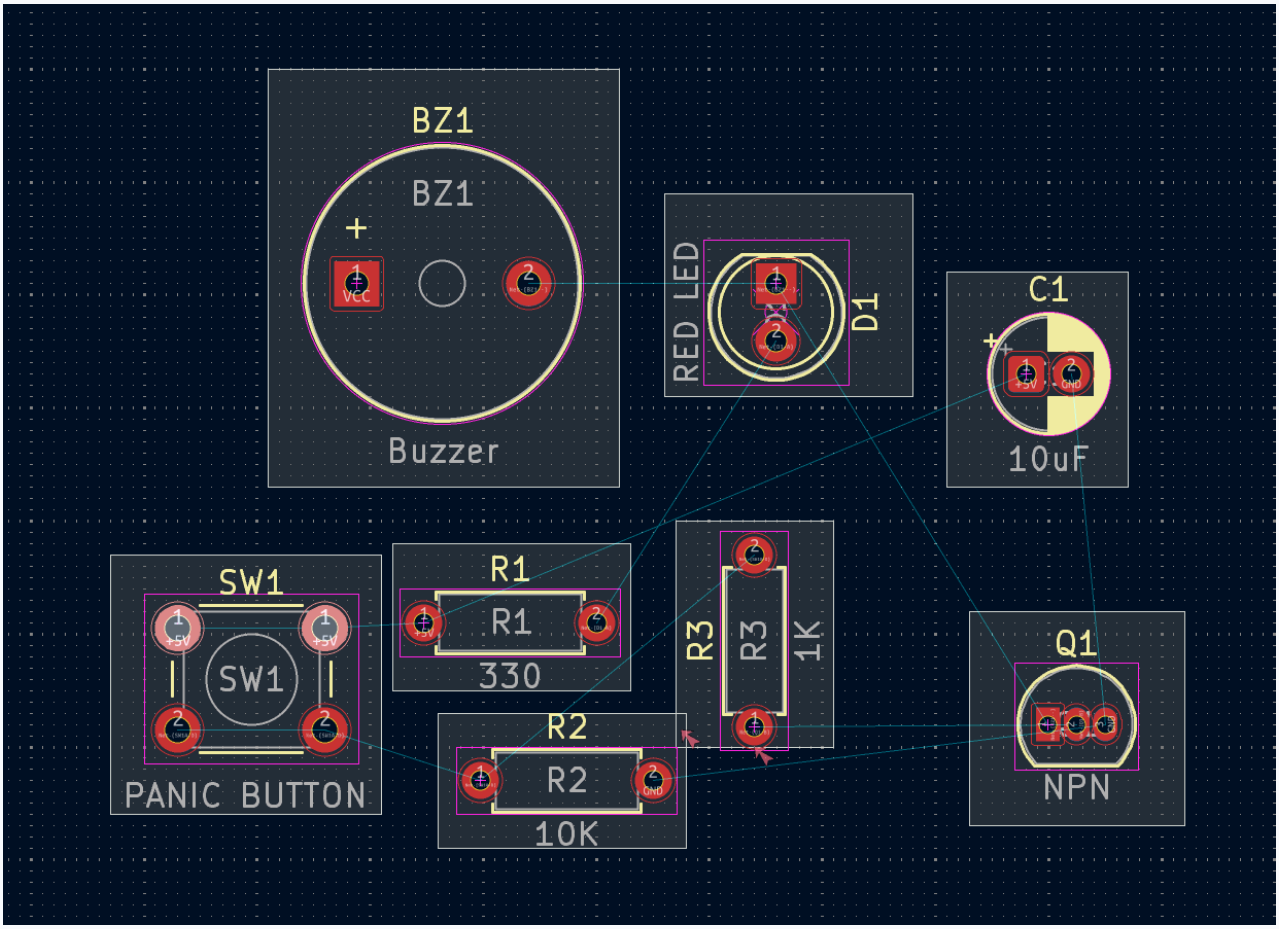
Mapping schematic symbols to physical hardware packages using KiCad's Footprint Assignment Tool:



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Symbol : Footprint Assignments
BZ1 -            Buzzer : Buzzer_Beeper:Buzzer_12x9.5RM7.6
 C1 -              10uF : Capacitor_THT:CP_Radial_D5.0mm_P2.00mm
 D1 -           RED LED : LED_THT:LED_D5.0mm
 Q1 -               NPN : Package_TO_SOT_THT:TO-92_Inline
 R1 -               330 : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
 R2 -               10K : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
 R3 -                1K : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
SW1 -      PANIC BUTTON : Button_Switch_THT:SW_PUSH_6mm_H4.3mm
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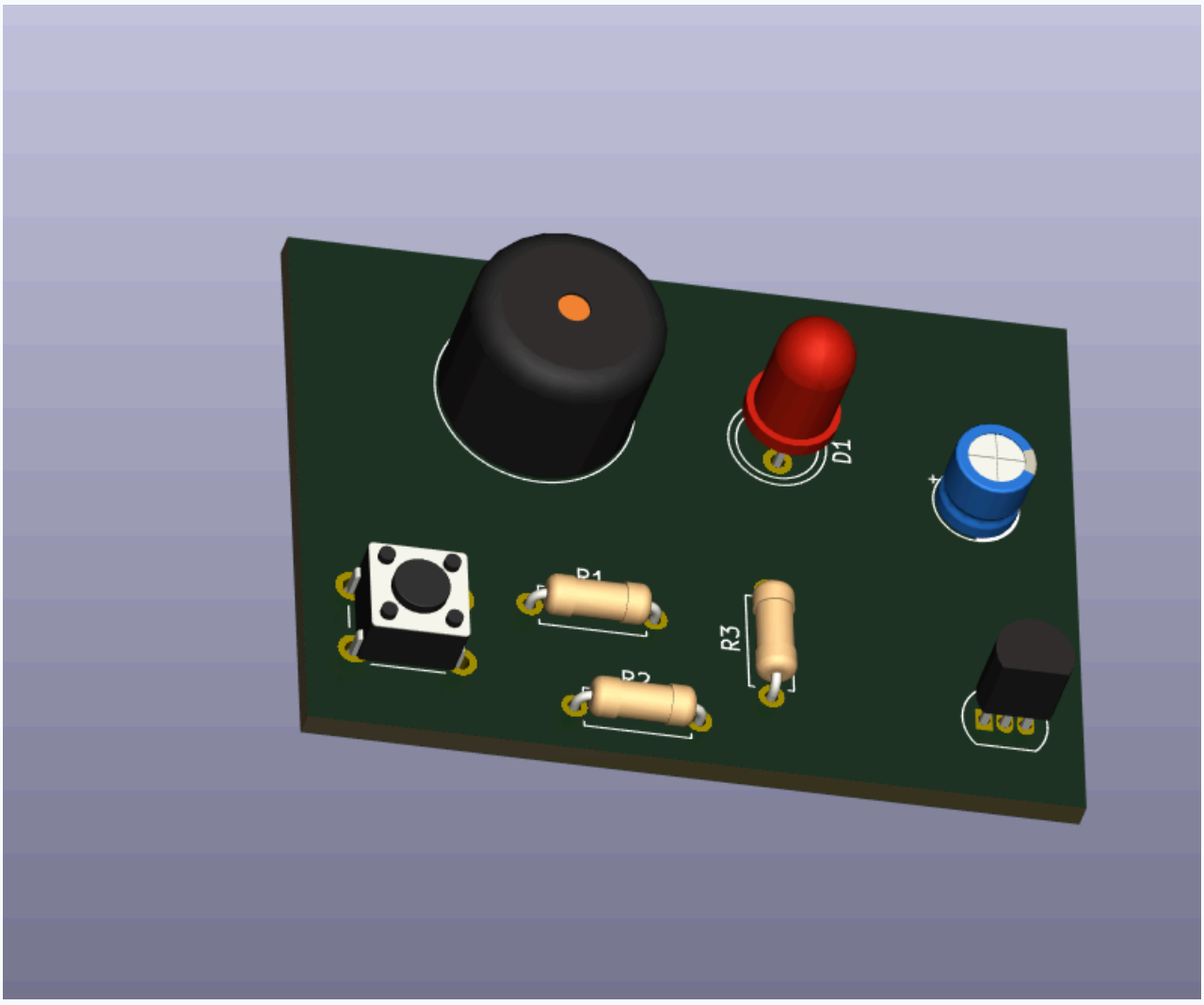
2D PCB Layout Design

Component placement, net ratlines, and footprint alignment in the 2D PCB Editor:



3D PCB Visualization

Ray-traced 3D model preview displaying physical component heights, orientation, and board proportions:



Bill of Materials & Design Analysis

1. Bill of Materials (BOM)

Designator	Value / Component	Assigned KiCad Footprint Package	Qty
BZ1	Buzzer	Buzzer_Beeper:Buzzer_12x9.5RM7.6	1
C1	10uF Capacitor	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm	1
D1	RED LED	LED_THT:LED_D5.0mm	1
Q1	NPN Transistor	Package_TO_SOT_THT:TO-92_Inline	1
R1	330 Ω Resistor	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1
R2	10K Ω Resistor	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1
R3	1K Ω Resistor	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1
SW1	Panic Push Button	Button_Switch_THT:SW_PUSH_6mm_H4.3mm	1

2. Key PCB Design Considerations

- **Trace Routing & Width:** Signal and power tracks are sized adequately to avoid impedance drops and handle current drawn by the buzzer and LED.
- **Component Clearance:** Through-hole components are positioned to maintain physical clearance for ease of manual soldering.
- **Polarity Verification:** Critical polarized elements (Capacitor C1, LED D1, and Buzzer BZ1) have clear silk-screen orientation markers on the PCB.

3. Conclusion

The Emergency Panic Alarm / Electronic Doorbell PCB design was successfully completed in KiCad E.D.A. verified through Design Rule Checks (DRC), footprint mapping, and 3D rendering. The finalized layout is ready for manufacturing output (Gerber and Drill files).